Simulating quantum spin ice in a small box: the winding-free campaign, explained from the beginning

Pedagogical companion to `twist_qsi_demo`

July 2026

Contents

1 Why this document exists

This is the story of a long computational campaign, told so that a reader with *no* mathematical background can follow every step. The campaign asks a deceptively simple question:

> *Can we trust a computer simulation of a quantum magnet when the simulation is forced to live in a tiny box?*

For the material family we care about — *quantum spin ice*, realized in cerium pyrochlore magnets such as $Ce_2Zr_2O_7$ and $Ce_2Hf_2O_7$ — the answer turned out to be "**no, not naively**": the tiny box invents a fake physical process that is *stronger* than the real one, and every naive simulation measures the fake. The campaign (i) identified the fake process precisely, (ii) proved that the popular fix does *not* work, (iii) constructed a fix that provably works, (iv) pushed that fix into the strongly interacting regime where the usual approximations collapse, (v) translated the whole construction into the language of emergent gauge theory, where each hard-won fact turns out to have an old name and an older proof (Sec. 10), and (vi) mapped out what is left to do — including what the corrected theory positively predicts for experiments on the cerium materials (Sec. 10.6). Along the way it collected several wrong turns that are as instructive as the successes; they are all documented here.

Every number quoted in this document was computed and cross-checked in the campaign; nothing is schematic except the drawings.

2 The stage: what quantum spin ice is

2.1 Spins on a pyrochlore lattice

The magnetic atoms in these materials sit on a *pyrochlore lattice*: a network of corner-sharing *tetrahedra* (a tetrahedron is a pyramid with four triangular faces and four corners). Each corner hosts one magnetic moment — a "spin" — which you can picture as a tiny arrow that is only allowed to point either *into* the tetrahedron or *out of* it.

The dominant interaction, called J_{zz}, is a simple rule of thumb:

each tetrahedron is happiest when exactly two of its four arrows point in and two point out.

This is the famous **ice rule**, named for its exact analogy with the positions of hydrogen atoms in ordinary water ice.

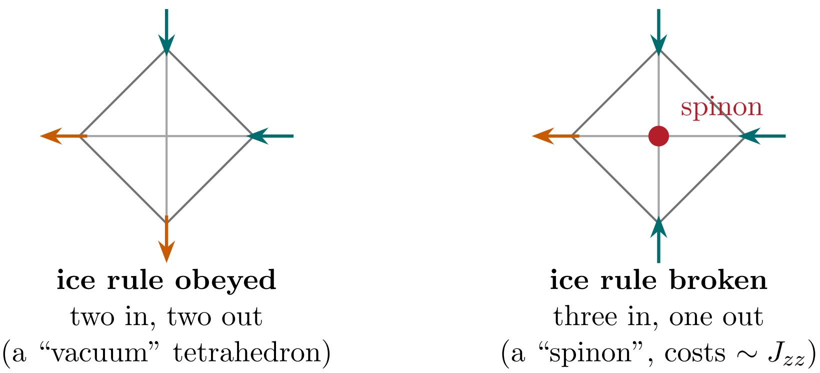


Figure 1: One tetrahedron of the pyrochlore lattice, seen from above. Each of the four corners carries a spin that points in (teal) or out (orange). Left: an ice-rule configuration. Right: flipping one spin creates a defect — a *spinon* — with an energy cost of order J_{zz}.

The ice rule does not pick a single winner. On a lattice of N spins there is an enormous number of different arrangements that all satisfy “two in, two out” on every tetrahedron — for our 16-site simulation cell there are exactly 90 of them, for the 32-site cell 2970. This collection of equally happy states is called the **ice manifold**. All the interesting low-temperature physics happens *inside* this collection.

2.2 Three characters emerge

Quantum mechanics lets the system tunnel between different ice configurations, and out of this tunneling emerges a spectacular fiction: the system behaves, at low energy, like a miniature universe with its own electromagnetism. Three characters matter for us:

Spinons. Break the ice rule on one tetrahedron (three-in–one-out) and you create a particle-like defect. Spinons are the “electric charges” of the emergent electromagnetism. They are expensive (energy $\sim J_{zz}$) and show up in measurements as a broad feature at temperatures $T \sim J_{zz}/4$.

The photon / ring exchange. Within the ice manifold, the cheapest quantum motion is a *ring exchange*: six spins arranged around a hexagonal loop flip together (in→out→in$\cdots$ becomes out→in→out$\cdots$). This collective six-spin flip is the matrix out of which the emergent “light” (the photon) is built. Its strength is tiny:

$$g_6 = \frac{12\,|J_\pm|^3}{J_{zz}^2}, \tag{1}$$

where $J_\pm$ is the quantum (spin-flipping) part of the interaction. The power *three* appears because the loop must be built in three elementary steps (more on this below).

Flux. Going around a hexagon, the quantum amplitudes can add up as if a magnetic flux threads the loop. For $J_\pm < 0$ (the sign relevant to the cerium materials) theory predicts π-*flux*: each hexagon behaves as if half a flux quantum pierces it. Detecting this flux is one of the big experimental goals.

In plain words

A magnet whose spins obey an ice rule hides a fictional universe: defects act like electric charges (spinons), collective six-spin flips act like light (photons), and hexagons can carry magnetic flux. The photon scale g_6 is minuscule — it involves $|J_\pm|^3$, the cube of an already small number — so any pollution of the simulation at a *lower* power of $J_\pm$ will drown it. Keep that sentence in mind; it is the whole story.

2.3 The Hamiltonian, for reference

For completeness (skip freely), the model is the XXZ pyrochlore Hamiltonian

$$H \;=\; J_{zz}\sum_{\langle ij\rangle} S_i^z S_j^z \;-\; J_\pm \sum_{\langle ij\rangle}\big(S_i^+ S_j^- + S_i^- S_j^+\big), \tag{2}$$

with $J_{zz} > 0$ fixed to 1 (it just sets the unit of energy) and the single dial $J_\pm$, typically a few percent of J_{zz} in real materials, negative for the cerium pyrochlores. S^z is the in/out arrow; $S^\pm$ flips it.

3 The instrument: exact diagonalization on a torus

3.1 Why small boxes

"Exact diagonalization" (ED) means: write down *every* possible spin configuration, build the full table of quantum amplitudes between them, and let the computer find the exact energy levels. No approximation whatsoever — that is its glory. Its curse is size: with N spins there are 2^N configurations. Sixteen spins give 65 536 states (easy); 32 spins give 4.3 billion (heroic); 48 spins is beyond any computer for the full problem. So ED lives in small boxes, and to avoid fake surfaces we wrap the box around: the right edge is glued to the left edge, the top to the bottom, front to back. A box glued this way is a **torus** — a doughnut (in three dimensions, a triple doughnut).

3.2 Two kinds of loops on a doughnut

On a doughnut there are two fundamentally different kinds of closed loops, and this distinction is the geometric heart of the entire campaign.

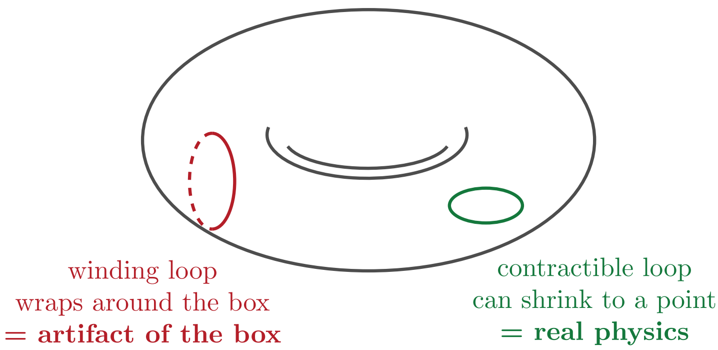


Figure 2: On a torus (a box with glued edges) a loop either shrinks to a point (green) or wraps around the box and cannot shrink (red). In an infinite crystal only the first kind exists. Everything the box invents lives on the second kind.

Contractible loops can be shrunk continuously to a point without ever leaving the surface. These loops exist in the infinite crystal too. The hexagonal ring exchange — the photon — lives on contractible loops.

Winding loops wrap around the doughnut. They exist *only because the box has been glued shut.* In the infinite crystal there is nothing they correspond to. Any process that lives on a winding loop is, by construction, an artifact of the simulation geometry.

The number of times a loop wraps around each of the three glued directions is called its **winding** $w = (w_x, w_y, w_z)$, three whole numbers. Contractible loops have $w = (0, 0, 0)$. A deep mathematical fact (we will meet it again in Section 6) says this triple of integers is the *only* topological label a loop on a torus carries — there is no finer distinction.

4 The villain: the box invents a process stronger than the physics

4.1 Counting steps

How strong is a collective flip process? In quantum mechanics, a process that requires k elementary spin flips, each of strength $J_\pm$, passing through intermediate states that cost energy $\sim J_{zz}$, has strength roughly

$$\frac{(\text{flip strength})^k}{(\text{energy barrier})^{k-1}} = \frac{|J_\pm|^k}{J_{zz}^{k-1}}.$$

Each flip is a "step up the ladder", and cheaper processes need fewer steps. The hexagon needs six spins flipped, done in three neutral steps (each step flips a $+/-$ pair), giving the $|J_\pm|^3$ of Eq. (1).

Here is the catastrophe. On the small torus there exist closed loops of only *four* sites that respect the ice rule — but every single one of them **wraps around the box**. (In the infinite pyrochlore lattice the shortest ice-respecting loop is the hexagon; the four-site loop only closes because the glued boundary gives it a shortcut.) A four-site loop needs just two neutral steps:

$$g_4 = \frac{4\,J_\pm^2}{J_{zz}}. \tag{3}$$

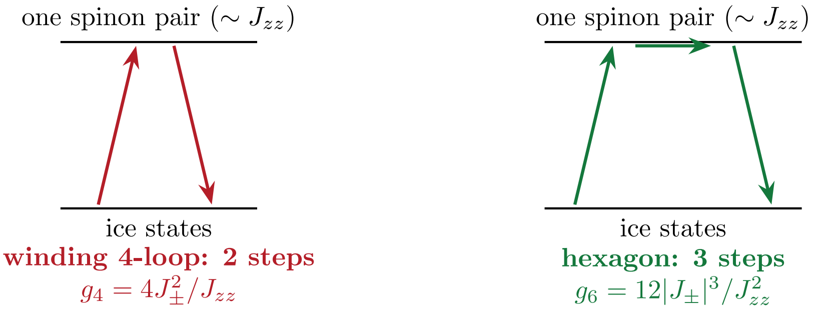


Figure 3: Why the artifact beats the physics. The fake wrap-around loop completes in two steps, the genuine hexagon needs three. Every extra step costs one more factor of the small number $|J_\pm|/J_{zz}$.

The ratio of fake to real is

$$\frac{g_4}{g_6} = \frac{J_{zz}}{3\,|J_\pm|}, \tag{4}$$

which *grows* as the coupling gets smaller — precisely in the physically relevant regime. At $J_\pm = -0.05$ the artifact is about seven times stronger than the photon; at $J_\pm = -0.01$, thirty-three times. And the box is full of these loops: the 16-site cell hosts 36 winding four-site loops and 48 boundary-wrapping hexagons, against only 16 genuine contractible hexagons.

4.2 The measured symptom

The clearest thermodynamic fingerprint of the photon sector is a bump (a "Schottky anomaly") in the specific heat $C(T)$ at a temperature set by the ring-exchange scale. The campaign measured, across the whole coupling range, where naive ED puts this bump:

	peak follows	measured exponent	ideal exponent
naive periodic ED	$g_4 \quad (T_{\text{peak}}/g_4 \approx 1.9 \to 1.5)$	$\lvert J_\pm\rvert^{1.99}$	2
winding-free ED	$g_6 \quad (T_{\text{peak}}/g_6 \to 1.02$ as $J_\pm \to 0)$	$\lvert J_\pm\rvert^{3.10}$	3

Measured against the photon scale g_6, the naive peak position wanders by a factor of *forty* across the scan. Naive ED does not measure the ring scale badly — it does not measure it *at all*; it measures the box.

In plain words

The simulation box invents a two-step shortcut process that only exists because the edges are glued together. The real photon is a three-step process. Since every step costs a factor of a small number, the fake two-step process dominates everything the simulation says about the photon: the low-temperature specific-heat bump, the flux, the dynamics. The fake even gets the *sign of the flux wrong*: the naive ground state shows zero-flux-like behaviour ($\langle\text{hexagon}\rangle = +0.116$) for both signs of $J_\pm$, where the true π-flux answer is negative.

5 First attempt: twist averaging — and why it fails

5.1 The idea

There is a classical trick for taming box effects, borrowed from electronic structure theory: *twisted boundary conditions*. When a spin flip crosses the glued edge, multiply its amplitude by a phase factor $e^{i\theta}$ — as if a magnetic flux θ were threaded through the hole of the doughnut. A process that wraps the box w times picks up the phase $w\,\theta$; a contractible process picks up nothing. So, the hope goes: **average over twists** and the wrapping processes, whose phases rotate around the clock, cancel to zero, while the physics is untouched.

The campaign's original claim (June 2026) was exactly this. **It is wrong**, and understanding *why* it is wrong was the first real discovery.

5.2 The failure, precisely

A ring-exchange process is not one path — it is the *sum of two* distinct virtual routes (which pair of spins flips first). The twist phase attaches to each *route*, not to the loop as a whole:

$$c(\theta) \;=\; -\frac{g_4}{2}\Big(e^{-iN_A\cdot\theta} + e^{-iN_B\cdot\theta}\Big), \qquad N_A + N_B = w.$$

Only the *sum* of the two route-phases equals the winding. The magnitude works out to $\lvert c(\theta)\rvert = g_4\,\lvert\cos(w\cdot\theta/2)\rvert$ — and here is the trap: on the 16-site cell, *every* winding four-loop has one route that never crosses the glued edge at all ($N_A = 0$, phase 1). Half of the fake amplitude is twist-blind.

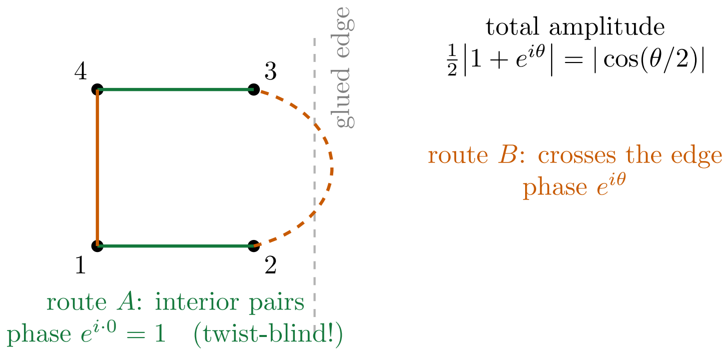


Figure 4: Why twist averaging cannot kill the wrap-around loop. The four-spin ring exchange is the sum of two virtual routes. Route B crosses the glued edge and feels the twist; route A uses only interior bonds and never sees it. No amount of averaging removes the twist-blind half.

The campaign then quantified every version of the idea:

- **Averaging measured quantities** (compute $C(T)$ at each twist, then average): weakest of all. A measured quantity depends on the *square* of amplitudes, and squares of cosines never average to zero.
- **Averaging over the special corner twists** $\theta \in \{0, \pi\}^3$: keeps exactly half of every fake coupling. Worse, a beautiful theorem (Theorem A below) shows these eight twists give *identical* spectra — they are the same physics in eight disguises, so averaging them at the level of observables does strictly nothing.
- **The best possible single twist**, $\theta = (\pi/2, \pi/2, \pi/2)$: kills 24 of the 36 fake loops exactly, but the other 12 survive at full strength. The surviving fraction of any twist-averaging scheme was proven to have a hard floor of $1/3$ — and because the surviving weight is *linear* in the choice of averaging weights, the optimum is a single point: no clever ensemble beats the best single twist.
- **The scaling test** (the damning one): with any observable-level twist scheme, the specific-heat peak still scales as $|J_\pm|^{1.8}$ — the artifact power — not the photon power 3. Twisting lowers the contamination; it never changes what is being measured.

Lesson learned (the hard way)

The original "theorem" that twist averaging removes winding contamination was retracted and the paper rewritten around the corrected theory. Two sub-lessons: (i) phases attach to virtual *routes*, not to the loop's topology — always compute the object you claim vanishes; (ii) the special twist (π, π, π), which naively flips the sign of every wrapping bond, was verified to leave the spectrum bit-for-bit unchanged — it is a *gauge transformation*, a pure relabeling. A sign flip of a matrix element is not interference.

6 The fix: a label that sees topology, and the mask

6.1 Transport: the bookkeeping number

If phases cannot kill the winding loops, we must find something that *measures* winding exactly, and delete by hand. The campaign found it, and it is beautifully simple.

Every flip process moves magnetization around. Take the set of spins the process flips; add up their positions with a + sign for raised spins and − for lowered ones. This gives a vector ρ, the **transport** (or polarization displacement) of the process. The campaign proved, by explicit enumeration over every loop of both simulation cells, the exact identity

$$\rho = \tfrac{1}{2}\, w \cdot L, \tag{5}$$

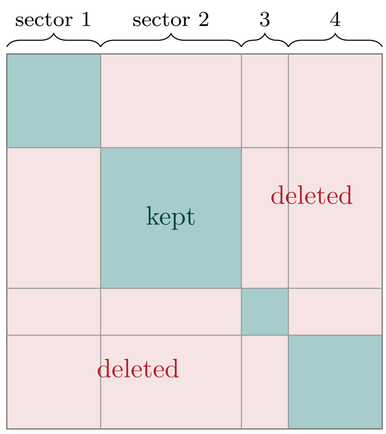
where w is the winding and L the box dimensions. Read it aloud: *transport is half the winding.* A contractible loop ($w = 0$) transports nothing — the raised and lowered spins balance perfectly. A winding loop *cannot* balance: it transports exactly half a box length. There is no overlap and no ambiguity: measuring ρ *is* measuring topology.

6.2 The mask

The recipe is now almost embarrassingly direct. Group the 90 ice states into families ("sectors") that share the same value of the polarization; on the 16-site cell there are 25 such sectors. Then, in the table of quantum amplitudes between ice states:

delete every entry that connects two different sectors.

That is the whole method. It is called the **mask**. By Eq. (5), an entry connecting different sectors is precisely a process with $w \neq 0$: a wrap-around artifact. An entry within a sector is a contractible process: untouched, to the last digit.



the amplitude table between ice states, sorted by transport sector

Figure 5: The mask. Sort ice states by their polarization sector; keep the diagonal blocks (contractible physics, green), delete everything off the blocks (wrap-around artifacts, red). Deleting can only remove processes — it can never invent new ones.

6.3 What the mask never asks: configurations, not eigenstates

A question that sounds central but that the mask never answers — because it never asks it — is: *which eigenstates of H are the ice band?*

Ice-ness, in this method, is a property of *configurations*, not of energy levels. A basis pattern either satisfies two-in–two-out on every tetrahedron or it does not — a yes/no check made pattern by pattern, tetrahedron by tetrahedron, involving no diagonalization, no energy threshold, and no coupling constant. The same 90 patterns qualify at $J_\pm = -0.03$ and at $J_\pm = -0.5$. The sectors and the mask inherit this: they are combinatorial objects, fixed before any physics is computed.

"Band" identity is then an *output label, never an input classification.* A winding-free band level is, by definition, a self-consistent root of an ice-sector block (the machinery of Secs. 7 and 8), and it arrives stamped with its sector because that is where it emerged. Even when such a root sits embedded inside the continuum (Sec. 9.7), there is no ambiguity about whether it is "band": it is a root of the ice-block map, full stop.

The contrast to draw is with the *frames* of Sec. 9 — constructions that answered the question the other way around, by hunting for the 90 most ice-*like eigenstates*. All of them die at strong coupling, and necessarily so: once bands mix, eigenstate identity is genuinely undefined. The mask survives because it staked its definition on something that cannot mix: the spin patterns themselves. (At strong coupling the physical states behind the roots are only $\sim 25\%$ ice by weight — the map does not care; the ice patterns say where a state is *anchored*, and the fold of Sec. 7 supplies everything else about it, exactly.)

6.4 Why this is safe, and why nothing finer exists

Three properties make the mask trustworthy:

1. **It only removes.** A projection deletes entries; it cannot create a process that was not there. So it cannot introduce new physics.

2. **It provably keeps the physics.** Contractible loops have $\rho = 0$ exactly (verified on every hexagon of both cells) and their amplitudes are untouched to machine precision.

3. **It is maximal.** The torus's mathematics (its "fundamental group" is $\mathbb{Z}^3$: loops are classified by three whole numbers, full stop) guarantees that winding is the *only* topological label. A process that winds $+1$ and then -1 (net zero) is topologically identical to a contractible one and is *correctly kept* — its counterpart exists in the infinite crystal too, as an ordinary, benign finite-size correction. The mask removes everything a topological label can remove, and nothing more. (Sec. 10 restates this maximality in the language of gauge theory, where it is a classical theorem about electric flux sectors rather than a feature of this method.)

6.5 Three theorems, in plain words

Later, the twist story and the mask story were united by three exact theorems (each verified to fifteen digits):

Theorem A (the eight disguises). The eight corner twists $\theta \in \{0, \pi\}^3$ produce *identical* energy spectra — each is the untwisted problem seen through a relabeling of states. Practical payoff: anything requiring "all eight corners" needs *one* computer run.

Theorem B (average = mask). The sophisticated "character average" over the eight corners — the operator-level averaging done right — is *exactly* the parity version of the mask. Verified entry by entry: zero mismatches out of 8100. The fancy procedure and the simple deletion are the same object, and the deletion is eight times cheaper.

Theorem C (the general dictionary). Any twist whatsoever factorizes into (a relabeling) $\times$ (a genuine flux). This explains every entry of the old "twists keep half" table and closes the subject: twists and masks are one theory, and the mask is its strongest element.

One more exact selection rule fell out as a bonus: within the winding-free world, the uniform magnetization response at zero wavevector vanishes *identically* (measured: 5×10^{-30}, versus 0.333 for naive ED). Every bit of that signal in a naive simulation is artifact.

In plain words

Count, for each quantum process, how far it shifts the magnetization pattern. Wrap-around processes shift it by half a box; real processes shift it by zero. Delete the shifters. That single sentence is the method, it costs almost nothing, it provably deletes all of the artifact and none of the physics, and mathematics guarantees nothing sharper is possible.

7 A toolkit interlude: folding the universe into a small matrix

Everything so far treated the ice manifold perturbatively — valid when $|J_{\pm}|$ is small. The campaign's second act pushes into *strong* coupling, and for that we need one more tool, the **Feshbach map** (also called a Schur complement). Despite the name it is one idea:

> *You care about a few states (the 90 ice states, call them P). The other 12 780 states (the spinon states, call them Q) matter only as temporary stopovers. Fold their effect into a corrected amplitude table for the 90, and solve the small problem.*

The folded table is

$$F(z) \;=\; \underbrace{PHP}_{\text{direct}} \;+\; \underbrace{PHQ\,\frac{1}{z - QHQ}\,QHP}_{\text{out to } Q \text{ and back}}, \tag{6}$$

where z is the energy at which you ask the question. The exact energies of the full problem below the spinon continuum are precisely the self-consistent solutions of $E = \operatorname{eig} F(E)$: the answer feeds back into the question. Nothing is approximated — F at the right z reproduces exact eigenvalues of the full H to machine precision.

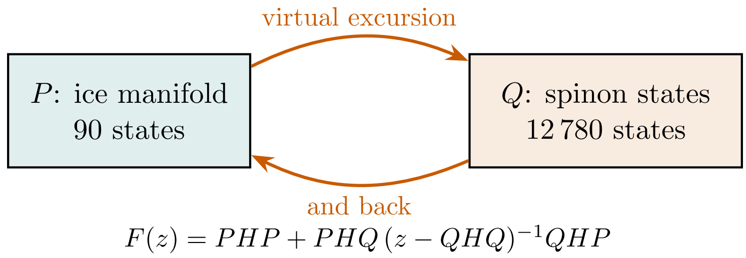


Figure 6: The Feshbach idea: excursions into the huge spinon space are summed exactly into a corrected 90×90 table. No frames, no overlaps, no approximation — and it exists at *any* coupling strength.

Two more characters:

Z, the ice weight (also: residue, purity). The fraction of a state that actually lives in P. It is a property of each exact eigenstate $|n\rangle$ of the full H, not of the map:

$$Z_n \;=\; \langle n|P|n\rangle \;=\; \big\|P|n\rangle\big\|^2 \;=\; \sum_{c\,\in\,\text{two-in–two-out}} |\langle c|n\rangle|^2 \;\in\; [0,1].$$

Be precise about what is being projected onto. The sum runs over the 90 *ice configurations* $|c\rangle$ — the spin patterns obeying the ice rule on every tetrahedron — not over Ising configurations in general: those form a complete basis, so projecting onto all of them is the identity and every Z_n would be 1. P is the rank-90 projector onto the span of that *subset*, $P = \sum_{c \in \text{two-in–two-out}} |c\rangle\langle c|$, diagonal in the Ising basis and purely combinatorial (Sec. 6.3); its complement Q is spanned by the other 12 780 configurations, each of which has at least one tetrahedron violating the rule, i.e. at least one spinon pair. So Z_n is the total probability that a measurement in the Ising basis returns an ice-obeying pattern. $Z_n = 1$ is a pure ice state, $Z_n = 0$ a state with no ice content at all, and $1 - Z_n$ is the *spinon dressing*: the weight the state has borrowed from Q by breaking ice rules virtually. One corollary worth flagging early: because the ice manifold is a set of configurations in the *local-z* frame, Z is defined relative to that frame — the same physical state has a different ice weight with respect to an x-basis ice manifold. Far from the Heisenberg point this is harmless (the z frame is the physical one); at it, it is the whole story (Sec. 9).

Two properties make Z more than a diagnostic. First, it is obtainable *without* eigenvectors: at a self-consistent root E_n of the branch $\lambda_n(z)$ of $F(z)$, $Z_n = [1 - \lambda_n'(E_n)]^{-1}$ — the flatter the folded map at a root, the purer the state (steep slope means the answer depended strongly on the asking-energy, i.e. on the excursions into Q). Second, it obeys an exact sum rule, $\sum_n Z_n = \operatorname{Tr} P = 90$, the sum running over *all* 12 870 eigenstates: the ice manifold's 90 dimensions are shared out among the whole spectrum, and the shares always total 90. Sec. 8.5 turns that sum rule into a thermodynamic identity. At weak coupling $Z \approx 0.99$ (states are almost pure ice); by $J_\pm = -0.30$, $Z \approx 0.25$ — states are mostly spinon dressing. $1 - Z$ controls every error and every convergence rate in what follows.

The mask, again. The transport label is defined on ice states, so the mask applies to $F(z)$ verbatim: delete entries of F between different sectors. This "masked Feshbach map" is the campaign's precision instrument: *winding-free and valid at any coupling.*

Worked example you can check by hand

A three-level miniature contains the whole method. Take states $|1\rangle, |2\rangle$ (our "ice states", in *different* transport sectors) and $|3\rangle$ (the "spinon"), with flip strength $t = 0.2$ into $|3\rangle$ and spinon cost $D = 1$:

$$H = \begin{pmatrix} 0 & 0 & t \\ 0 & 0 & t \\ t & t & D \end{pmatrix}.$$

Fold: excursions $1 \to 3 \to 2$ give $F(z) = \begin{pmatrix} f & f \\ f & f \end{pmatrix}$ with $f = -t^2/(D - z)$. The *off-diagonal* f connects different sectors: it is the analogue of a winding amplitude, generated purely by passing through the spinon. **Solve self-consistently:** $z = 2f(z)$ gives $z_0 = \frac{1}{2}\big(D - \sqrt{D^2 + 8t^2}\big) = -0.074456$, which is exactly the ground state of the full 3×3 matrix — fold-and-solve is exact. **Mask:** delete the off-diagonal; each sector alone solves $z = f(z)$, giving $z = \frac{1}{2}\big(1 - \sqrt{1.16}\big) = -0.038517$, twice (exactly degenerate) — the winding-free answer. Note it is roughly *half* the raw binding: on the real cluster, too, about half of the naive ground-state depth is artifact. **Counterterm** (Section 8.4): add $+0.037228$ to cancel the off-diagonal at z_0; the resulting honest Hamiltonian gives levels -0.039716 and -0.037228 — centred on the exact -0.038517 but split by 0.0025. That split is the "residual z-dependence": the counterterm cancels the artifact perfectly at one energy and imperfectly nearby. The purity here is $Z = 0.9352$; the split grows as $1 - Z$. Every phenomenon of the real campaign appears in this 3×3 example.

8 The method, settled: mask the fold

Everything up to here was a derivation. Twists were tried and shown to be incapable of removing the artifact (Sec. 5); the transport label was found to measure winding exactly (Sec. 6); the Feshbach fold was introduced to make the ice manifold's amplitudes explicit at any coupling (Sec. 7). Those three strands close here, into one instrument that the rest of this document simply *uses*.

The method is: build the fold, delete every entry between transport sectors, solve.

8.1 The protocol, in six steps

1. **List the ice configurations.** Every spin pattern with two-in–two-out on all tetrahedra: 90 on cubic-16, 2970 on FCC-32. This is a combinatorial list, fixed before any coupling is chosen (Sec. 6.3). It defines the projector P; $Q = 1 - P$ is everything else.

2. **Label them by transport.** For each configuration, the polarization ρ; group configurations with equal ρ into *sectors* (25 on cubic-16, 85 on FCC-32). Also combinatorial — integers, computed once.

3. **Fold.** At an asking-energy z, form $F(z) = PHP + PHQ\,(z - QHQ)^{-1}QHP$. This is exact: no expansion, no small parameter.

4. **Mask.** Delete every entry of $F(z)$ joining different sectors. Write the survivor $\Delta[F(z)]$. It is block diagonal, one block per sector.

5. **Solve self-consistently.** Find every E with $E = \operatorname{eig} \Delta[F(E)]$, branch by branch, with a genuine root finder (the fixed-point iteration converges at rate $1 - Z$ and will lie to you — see the lesson box in Sec. 9).

6. **Report.** The roots are exact winding-free energies. Each arrives stamped with its sector, with its ice weight $Z_n = [1 - \lambda_n'(E_n)]^{-1}$, and with an eigenvector inside the ice manifold, from which any observable that closes on ice configurations — the hexagon flux above all — is read off, averaged over the ground multiplet.

Steps 1, 2 and 4 involve no physics input whatsoever. Step 3 is the only expensive one, and it is one eigendecomposition of QHQ per parameter point. That is the whole algorithm.

8.2 Why the mask acts on the fold and never on H

The natural first guess — take H, find the matrix elements between ice states that wind, delete them — fails for a startling reason: **there are no such matrix elements**. The ice block of the bare Hamiltonian, PHP, is *diagonal.* A single pair flip acting on an ice state always breaks the ice rule on the tetrahedra it touches: it creates spinons and lands in Q. No ice configuration connects directly to any other.

Both the physical hexagon amplitude g_6 *and* the winding artifact g_4 are therefore *emergent*: assembled from multi-step paths that leave the ice manifold, wander through spinon space, and return. They live inside the resolvent $(z - QHQ)^{-1}$, smeared over the geometry of Q, not in any entry of H a scalpel could reach. The 3×3 miniature of Sec. 7 shows it nakedly: the offending amplitude is $-t^2/(D - z)$, and the original matrix contains only t and D.

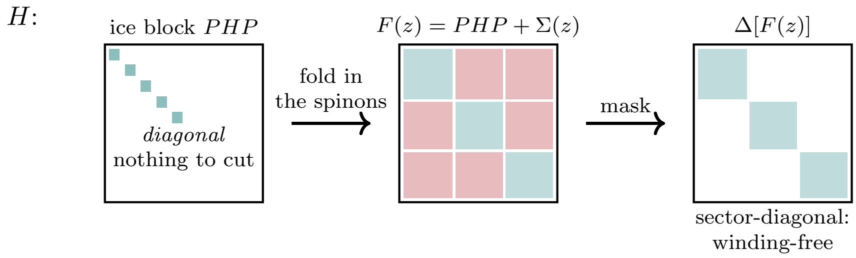


Figure 7: Where the mask acts. In H the ice block is diagonal, so there is literally nothing between sectors to delete. Folding the spinon excursions in *generates* the off-sector amplitudes (red) — those *are* the wrap-around processes — and deleting them from $F(z)$ is the mask. Teal blocks are the transport sectors that survive.

8.3 What it returns, and what it refuses to return

Being precise about the output is what keeps the rest of this document honest.

the masked map gives	**it does not give**
exact winding-free energies of every ice-descended level, at any coupling	anything about states with no ice content — the spinon continuum is outside its jurisdiction
the sector (topological) label of each level, for free	full-space wavefunctions; hence no ice-violation expectation, no defect density
the ice weight Z_n of each level, without eigenvectors	thermodynamics of the whole spectrum: the band alone has no upper Schottky peak
expectation values of any operator that maps ice to ice — the hexagon flux, the four-loop amplitude	an answer where no branch has a root: it returns *nothing*, and saying so is a result

The right-hand column is not a list of defects. Two of its four entries are the mask working correctly: it does not touch the spinons (it must not — they are physics), and it refuses to invent a band where none survives.

8.4 The one honest extension, and why it can only be a convention

For full thermodynamics — both peaks of $C(T)$, the entropy, expectation values on the true dressed ground state — one wants a single Hermitian operator on the whole 2^N space that is winding-free. **No such operator exists**, and the reason is exact.

The transport label is defined on ice configurations only. A single spinon hop moves the polarization by a *quarter* of a box period, so the label is not even defined on Q, let alone conserved there. (This is not a failure of imagination; Sec. 10.2 identifies it with a known theorem of gauge theory — dynamical charges destroy flux superselection — so no cleverer bookkeeping can exist.) Any full-space construction must therefore make an arbitrary choice about the $1-Z$ of each state that lives outside the ice manifold. *Winding-free thermodynamics is a convention of order* $1-Z$ — and since Z is measured, the size of the convention is measured too.

The cheapest such convention is the **counterterm**: evaluate the masked-out part of the fold at the self-consistent ground energy z_0 and subtract it from H inside the ice block,

$$H + C, \qquad C = -(1-\Delta)\,F(z_0).$$

Because C touches only the 90×90 ice block, PHQ, QHP and QHQ survive byte for byte — spinons, dressing and the upper band still flow through the untouched couplings — and one ordinary diagonalization returns the entire spectrum. Its price is exactly the obstruction, made visible: $H + C$ reproduces the exact masked roots to 0.4% of the bandwidth at $J_\pm = -0.02$, 1.5% at -0.05, 7% at -0.10, 17% at -0.20 and 22% at -0.30 — the 3×3 miniature's level split, writ large.

8.5 Three definitions of one non-unique reference

Three constructions extend the masked band to the full spectrum. They agree exactly at weak coupling and diverge at strong coupling by precisely the amount the obstruction theorem demands.

The counterterm $H + C$. One solve, complete spectrum, residue $1 - Z$.

The splice. Replace the lowest 90 levels of the $H + C$ spectrum by the exact masked roots. Clean while the band lies wholly below the continuum ($|J_\pm| \lesssim 0.22$ on cubic-16); undefined once band and continuum interleave, because "the lowest 90 levels" stops being the band.

The two-sided fold. Fold the *other* way as well, $G(z) = QHQ + QHP\,(z - PHP)^{-1}PHQ$ — cheap, because PHP is diagonal (Sec. 8.2). Every exact eigenstate is then seen by both maps, with weights Z_n and $1 - Z_n$ that sum to one. Use that ice weight as an *identifier*: rank the raw levels by Z_n, remove the ninety highest *wherever they sit* (interleaved with the continuum included), and insert the ninety masked roots at weight one. No anchor energy, no positional assignment, and BICs slot in wherever they lie.

Lesson learned (the hard way)

The hijacked ensemble. The obvious version of the two-sided identity — keep both sums, but with masked roots carrying weight Z_n — was built and it fails spectacularly. At $J_\pm = -0.05$, where every definition must agree, it put *zero* heat capacity at the ring peak (splice and $H + C$: 0.157) and grew a spurious bump of height 0.65 at the artifact scale, taller than raw ED's own. The mechanism deserves engraving: the raw band levels enter with small weight $1 - Z_n \approx 0.06$ but at their *raw absolute energies*, a full artifact binding $\sim g_4$ *below* the masked band — the depth the mask exists to remove. At $T \sim g_6 \ll g_4$ the factor $e^{+\beta\Delta}$ beats any weight. **Weights cannot fix positions**: a substitution scheme must *remove* the levels it replaces, not down-weight them.

Two free gates come with the construction and are worth stating here, because they are checked at every parameter point before anything is believed: the trace identity $\sum_n Z_n = \mathrm{Tr}\, P = 90$, and the bimodality of the ice weights that makes the Z-ranking sharp (band 0.89–0.97 against continuum ≤ 0.083 at weak coupling). How far the three definitions actually diverge, and where, is measured in Sec. 9.

In plain words

One instrument, one convention, one error bar. The mask on the folded map is exact and needs no apology: it answers questions about the ice-descended band at any coupling. The moment you ask a question about the *whole* spectrum — a heat capacity, an entropy — you are forced to decide what "winding-free" means for states that are half spinon, and mathematics guarantees that decision is not unique. So we make it three different ways and quote how much they disagree. At weak coupling: not at all, to the digit. By $J_\pm = -0.40$: by a factor of three.

8.6 What the mask is not

Three clarifications that repeatedly saved time.

It is **not a twist average**. It is what the twist average would have been if twist averaging worked: Theorem B of Sec. 6.5 says the corner character average *equals* the parity mask entry for entry (zero mismatches out of 8100), and the mask is eight times cheaper. Averaging survives as the derivation and as an oracle, never as the algorithm.

It is **not a truncation**. Nothing is dropped from the Hilbert space; QHQ enters $F(z)$ in full. (On clusters too large for a dense complement a truncation *is* imposed on top, by necessity, and then its convergence must be laddered and quoted — Sec. 9.9.)

It is **not a fit**. There is no parameter to tune, no window to choose, no reference state to select. Every ingredient is either combinatorial (the configurations, the labels, the sectors) or exact linear algebra (the fold, the deletion, the roots).

9 The XXZ line, measured

Everything up to here was construction. This section is the result: what exact diagonalization of the XXZ pyrochlore Hamiltonian

$$H \;=\; J_{zz}\sum_{\langle ij\rangle} S_i^z S_j^z \;-\; J_\pm \sum_{\langle ij\rangle}\big(S_i^+ S_j^- + S_i^- S_j^+\big), \qquad J_{zz}=1,$$

says once the mask is applied, coupling by coupling, from the perturbative regime down to $J_\pm = -0.5$.

Six independent calculations feed it, all on the cubic-16 cluster unless a row says otherwise: (i) the microscopic amplitude audit, which compares matrix elements against the perturbative predictions; (ii) raw ED of the complete 65 536-state spectrum; (iii) the counterterm spectrum $H+C$, same size; (iv) the exact masked band from self-consistent roots of $\Delta[F(z)]$; (v) the anatomy of the masked map itself, amplitude class by amplitude class; and (vi) the resonance calculation that asks whether any of these levels has a decay width. Where two of them measure the same thing, the agreement is quoted. Nothing below is fitted or modelled.

9.1 First, the artifact is real, and the mask removes it exactly

Before any physics, the premise: the box *does* invent a four-site process, it *is* the dominant one, and deletion *does* leave the hexagon untouched. All three are matrix-element statements, checkable at a coupling small enough that perturbation theory is exact.

At $J_\pm = \mp 0.005$, comparing the ice-block amplitudes of the raw and winding-free operators against $g_4 = 4J_\pm^2/J_{zz}$ and $g_6 = 12|J_\pm|^3/J_{zz}^2$:

	$J_\pm = -0.005$	$J_\pm = +0.005$
raw winding four-loop element, in units of g_4	0.970	1.030
raw hexagon element, in units of g_6	0.971	1.028
ratio (artifact / physics)	**66.6**	**66.8**
winding-free four-loop element (largest)	**0**	**0**
winding-free hexagon element, in units of g_6	0.971	1.028
spread over the 16 hexagons	7×10^{-16}	1×10^{-15}
largest surviving non-hexagon element	1.7×10^{-8}	1.5×10^{-8}

The perturbative formulas are confirmed to 3% at this coupling; the artifact is **sixty-seven times** the physics it is drowning; the mask sets it to zero — not “to 10^{-16}”, but to the floating-point number 0, because the deleted entries are never added; and the hexagon amplitude it must not touch is unchanged to the last bit, uniformly across all sixteen contractible hexagons.

A second audit, at the working coupling $J_\pm = +0.046$, adds the piece that matters for a torus. The cubic cell has 16 contractible hexagons, 48 *wrapping ones*, and 36 winding four-loops. In the raw operator the wrapping hexagons carry *exactly the same* amplitude as the contractible ones (0.00131421 against 0.00131421, agreeing to fifteen digits) — no local or energetic criterion could ever tell them apart, which is precisely why the label has to be topological. After the mask: contractible 0.00131421 (unchanged), wrapping 1×10^{-36}, winding four-loops 4×10^{-19}.

In plain words

The box's fake process is not a small correction to be estimated — it is sixty-seven times larger than the effect being measured, and the fake six-spin loops are numerically indistinguishable from the real ones by anything except topology. The mask deletes them completely and leaves the real ones bit for bit identical. Every number in the rest of this section rests on those two sentences being literally true, which is why they were checked first.

9.2 The temperature scale: where the specific-heat peak really is

The photon's thermodynamic fingerprint is the low-temperature Schottky anomaly in $C(T)$. Below, its position on a 12 000-point logarithmic grid with parabolic refinement, for the raw periodic spectrum and for the winding-free one, alongside the two candidate scales. ($T_{\rm ring}$ is the winding-free gauge peak, $T_{\rm per}$ the raw one; both are full spectra, so no ensemble mismatch enters.)

$J_\pm$	g_6	$T_{\rm ring}$	$T_{\rm ring}/g_6$	$T_{\rm per}$	$T_{\rm per}/g_4$	$T_{\rm per}/g_6$
−0.01	0.000012	0.000012	1.024	0.00075	1.866	62.2
−0.02	0.000096	0.000094	0.978	0.00271	1.691	28.2
−0.03	0.000324	0.000302	0.932	0.00554	1.539	17.1
−0.05	0.001500	0.001259	0.839	0.01295	1.295	8.63
−0.07	0.004116	0.003076	0.747	0.02179	1.112	5.29
−0.09	0.008748	0.005759	0.658	0.03148	0.972	3.60
−0.11	0.015972	0.009169	0.574	0.04171	0.862	2.61
−0.15	0.040500	0.017327	0.428	0.06355	0.706	1.57
−0.20	0.096000	0.028523	0.297	0.10094	0.631	1.05
−0.25	0.187500	0.039971	0.213	0.14225	0.569	0.76
−0.28	0.263424	0.048161	0.183	0.15603	0.498	0.59
+0.01	0.000012	0.000015	1.245	0.00091	2.279	76.0
+0.03	0.000324	0.000554	1.710	0.00980	2.722	30.3
+0.046	0.001168	0.002650	2.268	0.02548	3.010	21.8

Four readings, in order of importance.

Raw ED measures the box. Its peak sits at 1–2 times g_4 throughout, drifting slowly, exactly as a two-step process should. Measured instead against the photon scale it wanders from $62\,g_6$ to $0.5\,g_6$ — a factor of 120 across the scan. A quantity that changes by two orders of magnitude relative to the thing it is supposed to measure is not a noisy measurement of it; it is a measurement of something else.

The winding-free peak converges onto the photon scale. $T_{\rm ring}/g_6 \to 1.02$ as $J_\pm \to 0$: the exact perturbative prediction, recovered where it must hold, with no adjustable anything. This is the single most important validation in the campaign, because it is a place where the answer was known in advance.

The power laws come out right, and their split is physics. Fitting $T_{\text{peak}} \propto |J_\pm|^n$ over $|J_\pm| \le 0.031$ gives $n = 3.099$ winding-free (ideal 3) against $n = 1.994$ raw (ideal 2). That combined fit uses both signs of $J_\pm$; taken separately the π-flux side gives $n = 2.92$ and the zero-flux side $n = 3.28$ (raw: 1.83 and 2.16). The split is not scatter — it is the flux asymmetry showing up in the exponent, and it is the reason the two sides must be quoted separately when one cares about the third digit.

Flux costs the photon a factor of 2.7. At the same $|J_\pm|$, the ratio T_{ring}/g_6 is 1.02 at -0.01 against 1.24 at $+0.01$, and 0.84 at -0.05 against 2.45 at $+0.05$. The π-flux ground state resonates more weakly than the zero-flux one, by a factor growing from 1.2 to 2.9 over that range. Naive ED cannot see any of this: its artifact is even in $J_\pm$, so it reports nearly the same number on both sides ($1.87\,g_4$ against $2.28\,g_4$ at $|J_\pm| = 0.01$, most of that difference being contamination of the physical part rather than the flux law).

9.3 The spinon peak, which must not move — and does not

A cleaning procedure that improved everything would be suspicious. Spinons live *outside* the ice manifold; the mask is a deletion *inside* it; so the upper anomaly of $C(T)$ must be untouched. From the two complete spectra:

$J_\pm$	raw T_{spinon}	winding-free T_{spinon}	
-0.02	0.2510	0.2510	identical
-0.05	0.2533	0.2533	identical
-0.07	0.2557	0.2557	identical
-0.09	0.2581	0.2605	0.9%
-0.15	0.2630	0.2914	11%
-0.30	0.1634	0.4602	the two raw peaks have merged

Four-digit agreement through $J_\pm = -0.07$, which is the regime where the statement is meaningful, and then a controlled divergence as the artifact scale g_4 grows into the spinon scale and raw ED's two anomalies stop being separate maxima at all. (The raw entry at -0.30 is not the spinon peak descending; it is the label following a merged feature. Branch identity has to be tracked by continuity once $g_4 \sim J_{zz}/4$.)

9.4 The band, exactly

Now the object the mask is really about: the ninety winding-free levels descended from the ice manifold, obtained as self-consistent roots of $\Delta[F(z)]$ and verified against the full spectrum to 3×10^{-13}.

$J_\pm$	below edge	recovered above	lost	band bottom	GS mult.	gap above GS	median Z
-0.05	90	0	0	-0.07700	6	0.002841	0.936
-0.15	90	0	0	-0.60543	6	0.015264	0.636
-0.20	90	0	0	-0.99128	6	0.013576	0.473
-0.22	90	0	0	-1.15982	6	0.010634	0.414
-0.25	90	0	0	-1.42542	**2**	0.001206	0.338
-0.28	77	13	0	-1.71304	2	0.013105	0.275
-0.30	77	13	0	-1.90952	2	0.020638	0.221
-0.35	77	13	0	-2.41277	2	0.031457	0.157
-0.40	71	19	0	-2.92835	2	0.039192	0.119
-0.50	59	30	**1**	-3.98273	2	0.049856	—

Completeness. The band is entirely below the two-spinon continuum through $J_\pm = -0.25$. From -0.28 levels begin to cross the edge — thirteen of them, then nineteen by -0.40, thirty by -0.50 — and they are *recovered above it*, not lost, by the per-sector argument of Sec. 9.7. Across the entire scan exactly one level, at -0.50, is not accounted for.

Purity. The median ice weight falls from 0.94 to 0.12. By $J_\pm = -0.4$ a typical band state is 88% spinon dressing — and yet its energy is still exact, because the fold never assumed otherwise. This is the single number to keep in mind when reading any statement about strong coupling: it is also the size of the convention in every full-space extension (Sec. 8.4).

The ground multiplet changes character at -0.25. Through -0.22 the lowest level is six-fold degenerate — one state from each of six eight-dimensional transport sectors, degenerate to all printed digits because the mask makes the sectors exactly equivalent. At -0.25 a doublet from the twelve-dimensional sector drops below it and the multiplicity becomes two, leaving a splitting of 0.0012. This is a crossing *between topological sectors*, not a symmetry breaking, and it is discussed in Sec. 9.9, where it turns out to belong to the cluster rather than to the model.

The internal gap. The column "gap above GS" is a genuine feature of the winding-free spectrum and it radiates a real Schottky anomaly: at -0.22 the exact masked gap is 0.010634 and the independently constructed $H + C$ gives 0.010730. At weak coupling the resulting bump sits at $0.79\, g_6$, indistinguishable from the ring anomaly at $0.84\, g_6$ — it hides inside it — and by -0.22 the two are a factor of seven apart.

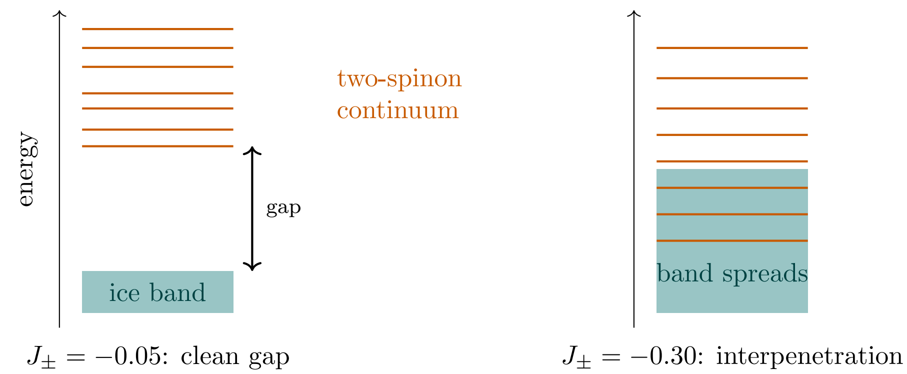


Figure 8: Measured behaviour, drawn schematically. At weak coupling the ice-derived band sits far below the spinon continuum. As $|J_\pm|$ grows the band widens (bandwidth $0.008 \to 0.35$ from -0.05 to -0.25) while the continuum descends, until they interpenetrate. Constructions that must *identify* band states die here; the masked fold, which never identifies anything, does not.

9.5 Inside the map: what the effective gauge theory becomes

The masked map is a 90×90 table whose every entry can be attributed to the spin pattern it flips. Classifying them at each coupling gives the effective theory directly, with no model assumed.

$J_\pm$	Z (ground)	$\lvert g_\text{eff}\rvert/g_6$	masked $\langle O_\text{hex}\rangle$	long/hex	$\lvert V/g\rvert$
−0.02	0.990	0.813	−1/4	0.134	0.007
−0.05	0.929	0.588	−1/4	0.317	0.011
−0.10	0.805	0.358	−1/4	0.567	0.006
−0.16	0.591	0.217	−1/4	0.787	0.004
−0.20	0.465	0.162	−1/4	0.898	0.010
−0.24	0.439	0.125	−1/4	0.989	0.014
−0.26	0.273	0.111	**0**	1.028	0.016
−0.30	0.222	0.089	0	1.096	0.019
−0.35	0.181	0.070	0	1.166	0.022
+0.046	0.890	1.264	+1/4	0.321	0.017

The photon amplitude is renormalized by more than an order of magnitude. g_eff is the hexagon amplitude the masked map *actually has* once every virtual spinon excursion is summed in. It is 81% of the perturbative g_6 at $J_\pm = -0.02$, half of it by -0.06, and 7% of it by -0.35. Quoting $12|J_\pm|^3/J_{zz}^2$ as "the photon scale" at $J_\pm = -0.3$ overestimates by a factor of eleven, and the measured peak position of the previous table tracks g_eff, not g_6. *This is the correct sense in which strong coupling is hard: not instability, renormalization.*

It stays a ring model — but stops being local. The column $|V/g|$ is the ratio of the diagonal (Rokhsar–Kivelson-like) term to the ring term. It never exceeds 0.022 anywhere on the line, so there is no diagonal takeover: the hypothesis that the band undergoes an RK-like transition inside the ice manifold is measured and dead. What does change is reach: "long/hex" compares amplitudes for flipping strings of eight or more spins to the hexagon amplitude, and it grows from 0.13 to 1.17. By -0.35 eight- and ten-spin loops resonate as strongly as hexagons. That is what "the effective theory breaks down" means here, quantitatively — a loss of locality, not a wall.

The flux is exactly quantized, and then exactly off. The masked ground multiplet's hexagon expectation is $-1/4$ to sixteen digits at every coupling through -0.24, and 0 to sixteen digits from -0.26: no drift, nothing in between. This is the π-flux label of the theory, recovered as a discrete number rather than a fitted contour, and it holds while the protected six-fold multiplet does.

9.6 The flux, and the sign raw ED cannot get right

The flux deserves its own comparison, because it is the sharpest failure of naive ED anywhere in the campaign. Measured on the full dressed ground state (so these are not the quantized ice-manifold numbers of the previous table but their full-space counterparts):

$J_\pm$	raw $\langle O_\text{hex}\rangle$	winding-free $\langle O_\text{hex}\rangle$	raw $\langle O_4\rangle$	verdict
−0.02	+0.115	−0.248	0.356	inverted
−0.05	+0.107	−0.238	0.348	inverted
−0.10	+0.091	−0.209	0.326	inverted
−0.20	+0.069	−0.149	0.290	inverted
−0.30	+0.058	−0.067	0.269	inverted
−0.35	+0.054	−0.009	0.261	inverted
+0.046	+0.119	+0.215	0.335	right by luck

Raw ED reports 0-flux at *every* coupling, including the entire region where the cerium materials live. The mechanism is one observation about powers: the artifact $g_4 = 4J_\pm^2/J_{zz}$ is **even** in $J_\pm$, the physical ring exchange $g_6 = 12|J_\pm|^3/J_{zz}^2$ is **odd**. For $J_\pm > 0$ they reinforce and the raw sign is accidentally right; for $J_\pm < 0$ they oppose, and since the artifact is the larger the measured sign is the artifact's. The middle columns show the same thing from the other side: the wrap-around amplitude $\langle O_4 \rangle \approx 0.33$ sits three to six times *above* the hexagon amplitude it is drowning, and collapses to 0.02–0.06 once the winding is removed.

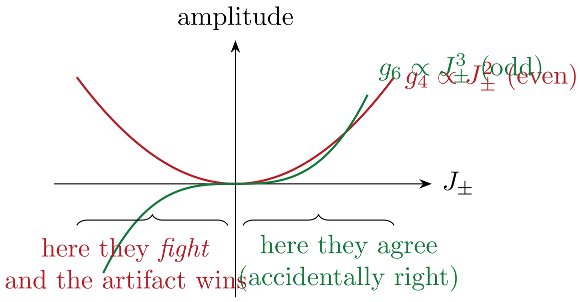


Figure 9: Why the naive flux is inverted on the cerium side. The artifact is even in $J_\pm$ and never changes sign; the real ring exchange is odd and does.

Lesson learned (the hard way)

An observable can be contaminated even when the state looks fine. The raw ice weight at $J_\pm = -0.10$ is a healthy 0.73 and nothing about that ground state looks broken. But flux is a *loop* observable, and the box's spurious loop is stronger than the real one, so the loop observable inherits the artifact's sign. Before trusting any quantity defined on a closed path in a periodic cluster, ask which closed path dominates it.

9.7 Levels leave the band — and none of them decays

From $J_\pm = -0.28$ the band is no longer wholly below the two-spinon continuum. The natural fear is that the photon is acquiring a lifetime and that the calculation is losing it. Both halves are measurably false.

The masked map is not one map but twenty-five: one independent block per transport sector, and *each block has its own continuum* — the spinon states that particular sector can reach. The quoted "continuum edge" is merely the lowest of these. So when a level of sector A rises above the global edge, the question is whose continuum surrounds it. If those states belong to *other* sectors, the level cannot decay into them — the transport label forbids it — and sector A's own map still has an ordinary real root there. The level is a **bound state in the continuum**. That is how the escaped levels are recovered: at -0.28 and -0.30 the thirteen missing ones include an entire twelve-fold multiplet sitting $0.16\,J_{zz}$ *above* the edge.

The decay widths, computed independently by golden rule on the winding-free poles and by continuing the self-consistent root into the complex plane:

$J_\pm$	roots found	largest golden-rule width	largest $\lvert\mathrm{Im}\,E\rvert$
−0.10	90	0 (underflow)	5×10^{-11}
−0.16	90	5×10^{-159}	8×10^{-11}
−0.20	90	3×10^{-46}	10×10^{-11}
−0.24	89	3×10^{-31}	6×10^{-11}
−0.28	77	3×10^{-15}	6×10^{-11}
−0.30	77	8×10^{-12}	7×10^{-11}

Zero, in other words, to the resolution of the calculation, at every coupling reached: complex iterations started off the real axis flow back onto it. **In the winding-free world the photon**

never decays. What naive ED shows as “the band merging into the continuum” is therefore a statement about the raw spectrum’s own contamination, not a lifetime. Whether the photon of the *infinite* lattice can decay — a question sixteen sites cannot pose, since a sixteen-site “continuum” is a comb of discrete lines — is answered by the field theory in Sec. 10.4, and the answer explains the measured zeros: below the two-spinon threshold the width vanishes kinematically.

foreign-sector continuum
level of sector A amid states
of sectors $B, C, \ldots$: decay forbidden,
$\Gamma = 0$ (measured: a BIC)

same-sector continuum
hybridization allowed: the root
turns complex, $E - i\Gamma/2$,
width Γ = a physical lifetime
(not reached on this cluster)

Figure 10: Two ways to sit inside a continuum. Left: the surrounding continuum belongs to other transport sectors; the per-sector root stays real and the state cannot decay — this is the measured case throughout the scan. Right: the continuum in the state’s *own* sector reaches it, the real root ceases to exist, and the honest object is a complex root whose imaginary part is a genuine decay rate — physics to report, not an artifact to repair. Which of the two occurs depends on the density of the spinon continuum, and on sixteen sites that density is itself a finite-size quantity.

9.8 The low-temperature structure, and how it was told apart

At strong coupling $C(T)$ grows extra structure *below* the ring anomaly. Asking whether it is physical turned out to have three answers, because “the third peak” named three different objects. The referee in each case is the exact masked spectrum: a degeneracy it declares exact must not radiate a bump, and a gap it certifies must.

Artifact. In the exact masked map the six sector ground states are degenerate with spread 0.00000. The counterterm splits them — it cancels transport inside the ice block at one energy while PHQ keeps transporting — by a fraction of the bandwidth growing from 0.011 at -0.10 to 0.115 at -0.30, and the split multiplet radiates a spurious Schottky bump ($T = 0.0013$ at -0.15, 0.006 at -0.20). Surgical proof: keeping only the five lowest $H + C$ levels reproduces it (0.00548 against 0.00564 at -0.30); collapsing their splitting removes it exactly. It is absent from the splice and the fold, whose lowest levels *are* the masked roots.

Physical, certified. The internal-gap Schottky of the band table: 0.010634 (masked) against 0.010730 ($H+C$) at -0.22, two independent constructions agreeing to three digits, hiding inside the ring anomaly at weak coupling and separating from it by a factor of seven by -0.22.

Physical, and cluster-specific. The $6 \to 2$ multiplicity change at -0.25 leaves an exact splitting of 0.0012 that radiates a maximum at $T \approx 4\times10^{-4}$ — a heat-capacity fingerprint

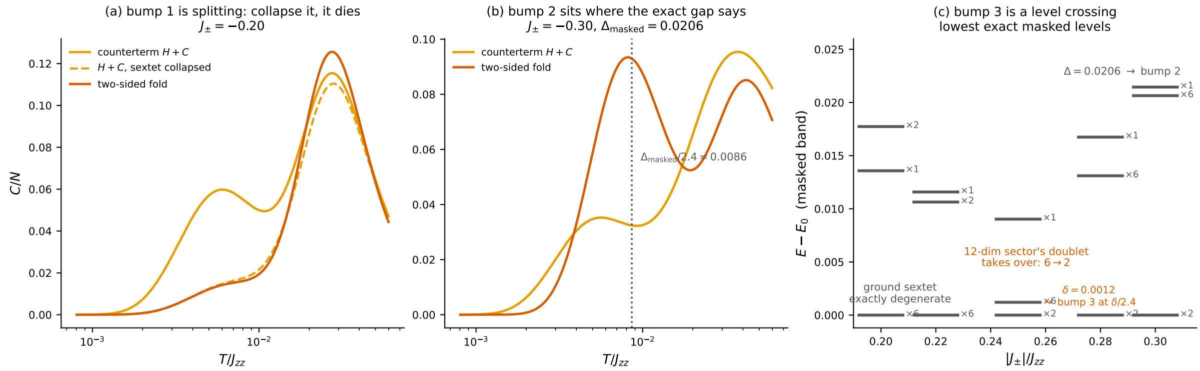


Figure 11: The taxonomy, demonstrated from the computed spectra. (a) At $J_\pm = -0.20$ the counterterm curve carries a low-temperature bump the fold lacks; collapsing the six lowest $H + C$ levels — the image of the exactly-degenerate masked ground sextet — to their mean (dashed) kills the bump and lands the curve on the fold's. (b) At $J_\pm = -0.30$ the fold peaks at the two-level Schottky position $\Delta/2.4 = 0.0086$ predicted by the exact masked internal gap $\Delta = 0.0206$. (c) The lowest exact masked levels versus coupling: the ground sextet is degenerate to all digits through -0.22; at -0.25 the twelve-dimensional sector's doublet dives below it, leaving $\delta = 0.0012$.

of ground-sector reorganization, visible without inspecting a single eigenvector. Whether it survives on a larger cluster is the subject of Sec. 9.9, and the answer is no.

In plain words

Same temperature window, three causes. A counterterm error split an exact degeneracy — fake, and diagnosed because the exact spectrum says those states are degenerate to all digits. A real gap inside the clean band — real, certified by two independent constructions agreeing to three digits. And a change in which sector owns the ground state — real on this cluster, and a bonus, since the specific heat announces the level crossing on its own. The rule is the same in all three: exact degeneracies must stay silent, certified gaps must speak.

9.9 How much of this is the cluster

Everything above is sixteen sites. The same line was then run on the 32-site FCC cluster with the identical protocol under a graded truncation of the spinon space (keep states within L pair-flip moves of the ice manifold; the L-ladder is calibrated on cubic-16, where $L = 4$ is the exact answer). Three results, and they are sobering in different directions.

The headline scale moves by a factor of three. At $J_\pm = -0.05$ the winding-free ring peak is $0.84\,g_6$ on cubic-16 and $0.28\,g_6$ on FCC-32. The protocol is identical, so the factor is the host. And it is largely *understood*: the peak position of a ring model is set by the hexagon-flip adjacency of the ice manifold, which is pure combinatorics — computable with no diagonalization at all — and those host coefficients are 1.070 (cubic-16) against 0.521 (FCC-32), already 0.49 of the measured 0.33. Cubic-16 is a poor host: 90 ice states in 25 sectors of dimension at most twelve, every state with 0 or 4 flippable hexagons and nothing in between, and a ground-state flux pinned by counting before any dynamics happens. FCC-32 has 2970 ice states, 85 sectors, the largest of dimension 396, flippabilities $0, 4, 6, 7, 8$, and no pinning.

The most exciting cubic-16 feature does not exist there. On FCC-32 the band bottom belongs to the unique zero-polarization sector at *every* coupling from -0.05 to -0.32, with the twelve 140-dimensional sectors forming an exactly degenerate runner-up and a gap that only grows ($0.0014 \to 0.0191$ at $L = 1$; $0.0018 \to 0.0573$ at $L = 2$). Deepening the truncation *entrenches* that ordering, whereas on cubic-16 deepening it is what *revealed* the crossing. So the $6 \to 2$ reorganization at -0.25 — and with it the little Schottky bump, and the switching-off of the quantized flux — is cubic-16 relaxing into the ordering FCC-32 has all along, not a physical transition. The coincidence with the theoretical terminus estimate $J_{\pm}/J_{zz} = -1/4$ was a coincidence.

The trend of the ring scale survives. The complete 2970-level FCC-32 band exists at thirteen couplings with no level escaping at any of them, and its ring peak declines smoothly from $0.281\, g_6$ at -0.05 to $0.0158\, g_6$ at -0.32 — the same qualitative story as cubic-16, one host coefficient lower. The truncation error is measured, not assumed: on cubic-16 the ring peak comes out $12\%/42\%/39\%$ low at $L = 1$ for $-0.05/-0.20/-0.30$, $0\%/11\%/19\%$ low at $L = 2$, and under 1% at $L = 3$. So FCC-32 numbers quoted at $L = 1$ are underestimates at strong coupling by up to $\sim 40\%$ if that error transfers; the weak-coupling factor of three between hosts is solid.

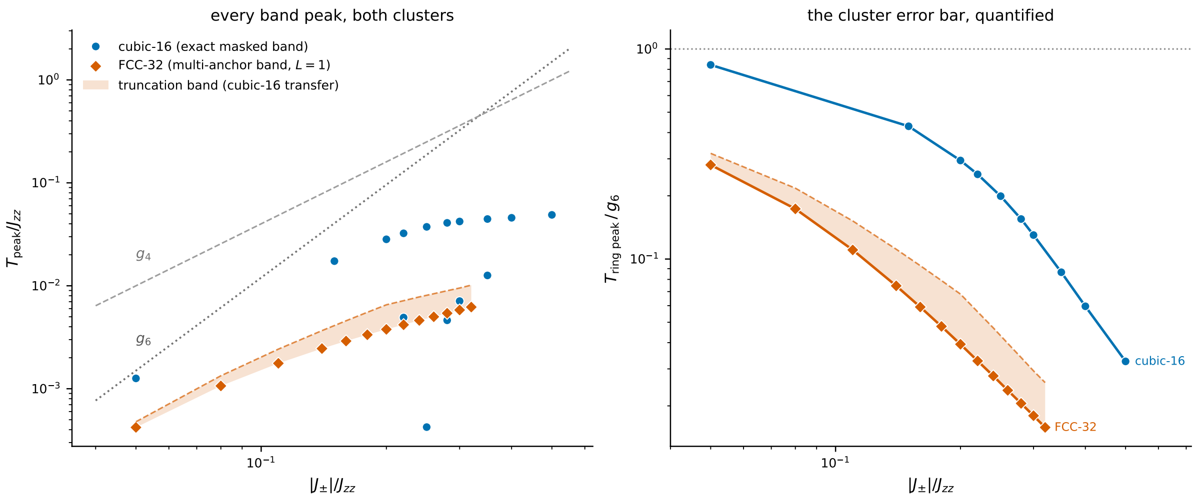


Figure 12: The winding-free ring peak on both clusters across the XXZ line. The protocol is identical; the hosts are not.

> **Lesson learned (the hard way)**
>
> **An instrument earns the big cluster by reproducing the small one.** A truncated FCC-32 pipeline already existed — one-pair truncation, map evaluated at a single anchor energy — and the obvious plan was to run it at the interesting coupling. Calibrated first on cubic-16, where the answer is known, it showed *no crossing at all* even at -0.30, with multiplet gaps wrong by a factor of four, because a fixed anchor misses the self-consistent $\sim Z$ renormalization of level spacings. Run uncalibrated it would have returned a confident, wrong verdict on precisely the question at stake.

9.10 Thermodynamics, and the size of the convention

The band alone has no upper Schottky peak, so any full $C(T)$ requires one of the three full-space definitions of Sec. 8.5. Along this line they were all three computed, and their disagreement is

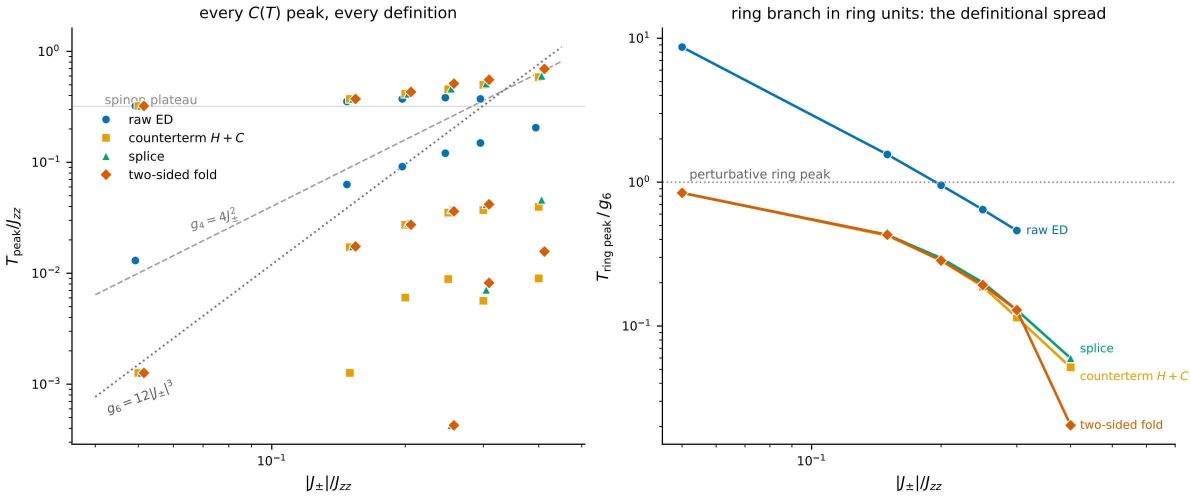


Figure 13: Every specific-heat maximum of every definition across the line (left) and the ring branch in units of g_6 (right). Three branches organize the left panel: the spinon anomaly on top, the ring branch below it, and the low-temperature structure of Sec. 9.8. Raw ED's ring branch runs along the artifact scale g_4 and crosses $1\,g_6$ near -0.22 purely by coincidence of scales. At -0.40 raw ED has no low-temperature peak at all — the artifact has grown into the spinon scale and swallowed every gauge feature — while the winding-free definitions still resolve structure.

the honest error bar.

$J_\pm$	counterterm	splice	two-sided fold
-0.05	0.00126	0.00126	0.00126
-0.15	0.01721	0.01744	0.01744
-0.20	0.02737	0.02848	0.02737
-0.30	0.03713	0.04183	0.04183
-0.40	0.03967	0.04590	0.01568

Identical to the digit at -0.05, within 12% through -0.30, and spread over a factor of three at -0.40. That growth is not a defect of any one definition: the obstruction theorem says "winding-free" is exactly defined only *on the band*, so every full-spectrum extension must decide which ninety levels the band replaces, and that decision is a convention of order $1 - Z$ — which the band table says is 0.88 at -0.40. The three constructions are three ways of making the same non-unique choice, and quoting their spread is the correct practice.

9.11 What was checked, and to how many digits

The numbers above are only worth their precision if the machinery around them was audited. For the record, on this line:

check	what it tests	result
masked roots vs. full spectrum	the fold-and-solve is exact	3×10^{-13}
$H + C$ ground vs. masked reference	the counterterm reproduces the exact ground root	$\leq 1.3 \times 10^{-13}$
$H + C$ ring peak vs. exact masked	where the convention still agrees with the exact object	$\leq 1.86\%$ (worst, at -0.22)
$H + C$ level-by-level vs. exact	where it does not: the $1 - Z$ residue, spread through the upper band	up to 19% of the bandwidth
$\sum_n Z_n = \mathrm{Tr}\, P$	the two-sided bookkeeping counts every level once	90 to ten digits
entropy $\int C/T\, \mathrm{d}T$	no level lost or invented, either definition	$\ln 2$ to 2×10^{-7}
hexagon amplitude, raw vs. masked	the mask does not touch physics	identical to 7×10^{-16}
winding amplitude after masking	the mask removes all of the artifact	exactly 0

Lesson learned (the hard way)

Three rulers that lied before the physics did. (i) The self-consistency $E = \mathrm{eig}\, F(E)$ solved by plain iteration converges at rate $1 - Z$; three rounds at strong coupling leave $\sim 10^{-2}$, which was misread for a week as a new physical effect complete with a spurious $C(T)$ peak. A secant step reaches 10^{-13} in 5–11 solves and the effect evaporated. (ii) Peak positions read off a 3000-point grid "agreed to 0.00%" — the grid step was 0.585%; the agreement was quantization. (iii) The entropy sum rule converges like $1/T_{\max}^2$, so truncating at $T_{\max} = 60$ leaves a deficit that looks exactly like a lost energy level and does not improve with grid refinement ($T_{\max} = 600$ resolves it). Audit the ruler before blaming the physics.

9.12 What the line establishes, and where it stops

Stated as plainly as the measurements allow:

- **Naive ED never measures the photon on this geometry.** Its low-temperature peak tracks g_4 with exponent 1.99, its flux has the wrong sign at every negative coupling, and its ice-block winding amplitude is $67\times$ the hexagon amplitude it hides.

- **The masked fold measures it, and is validated where the answer was known:** peak $\to 1.02\, g_6$ and exponent 3.10 as $J_\pm \to 0$, flux exactly $\pm 1/4$, hexagon amplitudes untouched to machine precision, spinon anomaly unchanged to four digits.

- **Strong coupling is renormalization, not instability.** The ring amplitude falls to $0.07\, g_6$ by -0.35 and the theory becomes longer-ranged (long/hex $0.13 \to 1.17$) with no diagonal takeover ($|V/g| \leq 0.022$); the band widens, the continuum descends, and thirteen levels cross the edge by -0.28 — every one of them with zero width. The raw ground state crosses nothing all the way to $J_\pm = -1$.

- **The one dramatic feature belonged to the box.** The $6 \to 2$ sector reorganization at -0.25, its Schottky bump and the accompanying loss of the quantized flux do not occur on FCC-32 at any coupling, and deeper truncation there entrenches the opposite ordering.

- **Two limits are structural, not technical.** Full-spectrum thermodynamics is a convention of size $1 - Z$ (measured: exact agreement at -0.05, a factor of three between definitions at -0.40); and the whole construction is anchored to the S^z ice manifold, so as $J_\pm \to -J_{zz}/2$ — the Heisenberg point, where the model is $J_{zz}\, \mathbf{S}_i \cdot \mathbf{S}_j$ and no spin axis is special — the mask built on z and the mask built on x become different operators with different spectra. The premise dissolves before the method does.

The frontier of this line is therefore not a failure of the instrument but a property of the hosts available: at -0.5 the ice description carries about a tenth of the low-energy weight, at -0.25 the sixteen-site cluster runs out of sectors, and the question of where the π-flux phase actually ends — somewhere in $[-0.25, -0.5]$, on the evidence here — needs the splitting between flux sectors measured against system size, which is a larger-cluster program rather than a better mask.

In plain words

On the XXZ line, in one paragraph. A sixteen-site periodic cluster invents a ring exchange sixty-seven times stronger than the real one, and every naive quantity that depends on a loop — the low-temperature specific heat, the flux — reports the invention instead: the peak sits at the artifact scale, and the flux comes out 0 where the true answer is π. Deleting the winding amplitudes from the folded map fixes both exactly: the peak converges onto the perturbative photon scale as the coupling goes to zero, the flux becomes a quantized $\pm 1/4$, the spinon physics is untouched, and nothing was invented anywhere. Pushed to strong coupling the photon does not die; it slows down by an order of magnitude, spreads over longer loops, and never acquires a decay width. What limits the answer in the end is not the artifact — that is gone — but the smallness of the box that hosts the calculation.

10 The gauge theory knew: the campaign restated in field-theory language

Everything in this document was derived microscopically: by enumerating loops, proving identities about polarization, folding matrices and measuring the results. The theory community has a second language for quantum spin ice — compact $U(1)$ lattice gauge theory with charged matter, the $\Phi^\dagger\Phi\, e^{iA}$ theory — in which every hard-won fact of this campaign turns out to have a name, and most of them have proofs that predate the material. Translating earns three things. It shows that the limitations of Sec. 8.4 are physics, not methodology. It imports tools for exactly the questions the small box cannot answer. And it turns several "sobering" measurements into positive predictions (Sec. 10.6).

10.1 The dictionary

The mapping is standard (Hermele, Fisher and Balents, 2004): the spin component S^z on a pyrochlore link is an electric field E, the raising operator S^+ is e^{iA}, and the two-in–two-out ice rule is Gauss's law $\nabla\cdot E = 0$. Everything in this document then has a second name:

this document	gauge theory
ice configuration	state obeying Gauss's law (vacuum sector)
spinon (3-in–1-out tetrahedron)	gauge charge Φ
the $J_\pm$ pair flip	matter hopping $\Phi^\dagger\Phi\, e^{iA}$: a charge moves and lays down electric string
hexagon ring exchange g_6	magnetic energy $\cos(\nabla \times A)$; its quanta are the photon
transport sector ρ	't Hooft electric flux sector through the torus cycles
winding four-loop g_4	a virtual charge pair created, wound once around a cycle, and annihilated: the flux-changing process
twist θ	background flat connection (Wilson line of an external gauge field)
corner twists $\{0, \pi\}^3$	the parity holonomies; Theorem C is the statement that any flat connection factorizes into gauge $\times$ holonomy
masked flux $\langle O_{\text{hex}}\rangle = -1/4$	the π-flux background of the emergent magnetic field

One entry of the dictionary carries a quantitative payoff immediately. The flux-changing process requires a charge to traverse a non-contractible loop of length ℓ_w, so its amplitude scales as $|J_\pm|^{\ell_w/2}/J_{zz}^{\ell_w/2-1}$: *relative to the hexagon it dies exponentially in system size*, and is anomalously strong only because $\ell_w = 4 < 6$ on the small cell. This single line reconciles the campaign with the literature: quantum Monte Carlo on large lattices (Shannon, Sikora, Pollmann, Penc and Fulde, 2012) measured the photon *without* any mask and was right to, because at large L the artifact is nonperturbatively small. The mask matters precisely where exact diagonalization lives, and the L-ladder of Sec. 9.9 is the crossover measured.

10.2 Why nothing finer than transport exists

The obstruction theorem of Sec. 8.4 — the transport label is exact on the ice manifold and *cannot* be extended to spinon states — looked like the method's private limitation. In gauge-theory language it is a classical theorem. In pure compact $U(1)$ on a torus the electric flux sectors are superselected ('t Hooft, 1979): no local process connects them. Add dynamical matter of unit charge and superselection is *destroyed* (the Fradkin–Shenker era, 1979): a Φ pair can wind a cycle and shift the flux by one, so no exact flux label survives on the full Hilbert space — for charge-q matter a $\mathbb{Z}_q$ remnant would survive, and $q = 1$ leaves nothing. Our spinons have unit charge. The quarter-period polarization step of a single spinon hop is this fact, microscopically.

The same translation absorbs the twist chapter. A charged state's flux depends on the Wilson string attached to it; only the combined winding of a pair's two strings is gauge invariant — which is exactly the route ambiguity $N_A + N_B = w$ that killed twist averaging in Sec. 5. And with unit-charge matter present, the only representation-independent projection a background holonomy can implement is the parity (center) one — which is Theorem B: of all conceivable character averages, precisely the $M = 2$ corner average survives, and it equals the parity mask.

Two consequences deserve to be said plainly. First, the $1 - Z$ convention of Sec. 8.5 is not a missed label waiting for a cleverer method; it is the measured size of a theorem. Any future scheme must pay it. Second, the search instinct "surely there is a deeper label underneath" has a definite answer: on the charge-free manifold, transport is the deepest label, and off it, there is none.

10.3 What the mask is: an emergent one-form symmetry, enforced by hand

Modern language sharpens the point once more. The conservation of electric flux through the torus cycles is the charge of a *one-form symmetry* (Gaiotto, Kapustin, Seiberg and Willett, 2015). In quantum spin ice this symmetry is *emergent*: explicitly broken by the spinon matter, exact below the spinon gap up to winding-tunneling corrections that vanish exponentially with L. The infinite lattice therefore enforces it dynamically. A four-site-circumference box does not — the tunneling g_4 is the symmetry-violating amplitude, resummed by the small torus into the dominant term. Under this light the whole campaign compresses to one sentence:

> *The mask enforces, at finite size and by hand, the emergent one-form symmetry that the thermodynamic limit would enforce dynamically.*

Its exactness on the band, its silence on the spinons, and its price of $1 - Z$ off the band are then three faces of "emergent, not exact."

This also situates the method in a lineage rather than presenting it as an ad-hoc repair. Quantum dimer models are the matter-*free* cousins of this problem — the dimer constraint is Gauss's law with the charges removed from the Hilbert space — and there, resolving torus spectra by winding sector has been standard practice since Rokhsar–Kivelson (1988) and Moessner–Sondhi (2001), because with no charges the sectors are exactly superselected and the "mask" is free. Quantum spin ice at finite J_{zz} puts the charges back. The masked fold is the controlled continuation of the dimer-model practice to that case: the fold restores the charge dynamics exactly (every virtual spinon excursion, to all orders), and the mask keeps the sector bookkeeping that the dimer limit had for free.

10.4 The photon's lifetime, where the box cannot go

Section 9.7 measured decay widths of zero — golden rule and complex continuation, every coupling reached — and noted that the honest failure mode, a level degenerate with its *own* sector's continuum, never occurs on this host. The field theory explains both halves and then answers the question the box cannot pose.

In the $\Phi^\dagger\Phi\, e^{iA}$ theory the photon width is the imaginary part of the transverse polarization bubble, $\Gamma(\omega) \propto \mathrm{Im}\,\Pi_T(\omega, k)$: the amplitude to convert one photon into a spinon–antispinon pair. That imaginary part vanishes *identically* below the two-spinon threshold, $\omega < 2\Delta_{\mathrm{spinon}}$. At weak coupling the photon lives at $\omega \sim g_6$, orders of magnitude below $2\Delta \sim J_{zz}$: the measured $\Gamma = 0$ is not a small number, it is a kinematic identity, and the winding-free photon of the infinite lattice is *exactly* stable there. A genuine width requires the spinon gap to close — the Higgs transition of the gauge–matter theory (Savary and Balents, 2012), where Φ condenses, the photon acquires a mass by the Anderson–Higgs mechanism, and critical fluctuations damp it. The lifetime question therefore has a sharp infinite-volume answer: *stable everywhere in the deconfined phase, damped only in a window around the Higgs transition.* Our measured continuum edge collapsing from $+0.67$ to $-1.63\,J_{zz}$ across the scan is the finite-size shadow of that closing gap, and the emergent fine-structure constant of spin ice being large (Pace, Morampudi, Moessner and Laumann, 2021) is why the approach to that window is so violent — it is the same strong photon–matter coupling that renormalizes g_{eff} down to $0.07\,g_6$ in the table of Sec. 9.

10.5 What is genuinely open

The electric label organizes everything above, and its usefulness fades exactly where Z collapses — the states become mostly matter, and the electric flux of a matter state is the thing the theorems forbid. The gauge theory owns a second, *magnetic* sector — monopoles of the emergent field, the π-flux background the campaign measures as the quantized $\langle O_{\mathrm{hex}} \rangle = -1/4$

— with its own one-form structure. Whether a magnetic label can organize the strong-coupling regime where the electric one goes soft is, to our knowledge, open; the quantized flux holding to sixteen digits while Z falls from 0.99 to 0.34 is at least a hint that something magnetic stays rigid after the electric bookkeeping has blurred.

10.6 What this offers experiment

Read as a referee report on naive simulation, this campaign is demolition: the peak was the box's, the flux sign was backwards, the one dramatic transition belonged to the cluster. Read as a corrected instrument, it makes positive, falsifiable statements — and they matter most for precisely the materials whose proposed couplings sit in the difficult regime.

The ring anomaly exists, but colder than advertised. The photon scale that reaches observables is g_{eff}, not the perturbative $12|J_\pm|^3/J_{zz}^2$: five times smaller near $J_\pm/J_{zz} = -0.18$ (the regime proposed for $Ce_2Hf_2O_7$) and eleven times smaller by -0.30. A specific-heat search that looked for the bump at the perturbative scale and found nothing has *not* falsified quantum spin ice; a bump found at the renormalized scale supports it. The corrected peak positions, coupling by coupling and on two hosts, are in Secs. 9 and 9.9 — with the honest host spread (a factor of three between cubic-16 and FCC-32) quoted as the finite-size error bar.

A discrete sign, not a fitted curve. For $J_\pm < 0$ the winding-free flux is π: quantized, $\langle O_{\text{hex}} \rangle = -1/4$ to machine precision, at every coupling where the protected multiplet survives. Naive theory got the sign backwards on the entire cerium side (Sec. 9); any probe sensitive to the flux sector tests the corrected prediction against the artifact one.

The photon is coherent. Below the two-spinon continuum its intrinsic width is exactly zero (Sec. 10.4); on the instrument's resolution that is a quality factor above 10^8. An observed gauge-photon linewidth is then a measurement of disorder, of instrument, or of proximity to the Higgs window — not an intrinsic property of the deconfined phase.

Where theory can promise numbers, and where only bands. The instrument itself says where its full-spectrum statements are sharp: for $|J_\pm|/J_{zz} \lesssim 0.22$ every construction agrees to a few percent, so a material fit is a genuine comparison; at -0.18 the whole 90-level band is intact (median ice weight 0.54) and quantitatively exact. Beyond -0.25, full-spectrum thermodynamics carries the measured convention spread and only band-resolved statements (energies, flux, sector structure) remain exact. A fit that lands there should be compared against band observables, not against $C(T)$ wholesale.

Old comparisons should be re-based, not discarded. Any material conclusion calibrated against small-cluster periodic ED inherited the artifact: peak positions tracking g_4, inverted flux, a spurious uniform response (0.333 naive against an exact 0). The winding-free tables of Sec. 9 are drop-in replacements at the same cost, while large-lattice QMC needed no correction and agrees with the corrected picture by the ℓ_w scaling of Sec. 10.1.

In plain words

The campaign found, spin by spin, what the gauge theory knew abstractly: the small box's fake process is a flux-sector tunneling that the infinite material forbids; no label finer than flux exists, and with charges around even flux is only emergent; the mask is that emergent conservation law, imposed by hand at finite size. The translation is not decoration. It proves the method's limits are nature's, it explains why big simulations never needed the fix, it answers the lifetime question the box cannot ask — the photon is exactly stable below the two-spinon threshold — and it converts a demolition into predictions: a colder ring anomaly, a quantized π-flux sign, a coherent photon, and a stated regime of validity for every number.

Starting points in the literature. Not a bibliography; one entry per idea used above. The gauge-theory mapping: M. Hermele, M. P. A. Fisher and L. Balents, Phys. Rev. B **69**, 064404 (2004). The photon seen numerically on large lattices: N. Shannon, O. Sikora, F. Pollmann, K. Penc and P. Fulde, Phys. Rev. Lett. **108**, 067204 (2012); O. Benton, O. Sikora and N. Shannon, Phys. Rev. B **86**, 075154 (2012). Gauge–matter mean-field theory and the Higgs transition: L. Savary and L. Balents, Phys. Rev. Lett. **108**, 037202 (2012). Electric flux sectors on the torus: G. 't Hooft, Nucl. Phys. B **153**, 141 (1979); their fate with dynamical matter: E. Fradkin and S. Shenker, Phys. Rev. D **19**, 3682 (1979). One-form (higher-form) symmetries: D. Gaiotto, A. Kapustin, N. Seiberg and B. Willett, JHEP **02** (2015) 172. The emergent fine-structure constant of spin ice: S. D. Pace, S. C. Morampudi, R. Moessner and C. R. Laumann, Phys. Rev. Lett. **127**, 117205 (2021). Winding sectors in the matter-free cousin: D. S. Rokhsar and S. A. Kivelson, Phys. Rev. Lett. **61**, 2376 (1988); R. Moessner and S. L. Sondhi, Phys. Rev. Lett. **86**, 1881 (2001). Reviews of quantum spin ice: M. J. P. Gingras and P. A. McClarty, Rep. Prog. Phys. **77**, 056501 (2014); L. Savary and L. Balents, Rep. Prog. Phys. **80**, 016502 (2017).